

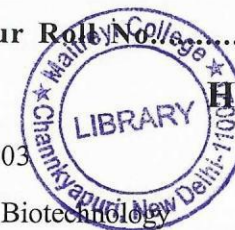
- (b) Explain calcium chloride method of transformation. (5)
6. (a) Explain how recombinant Insulin is produced using recombinant DNA technology. (7)
- (b) Describe the use of TALENS as a tool in gene editing. (5)
7. Write short notes on any **three** of the following : (4×3=12)
- (a) Molecular diagnosis of Sickle Cell Anaemia
- (b) Production of pharmaceuticals in transgenic animals
- (c) Microarray
- (d) Bacteriophage lambda
- (e) Scope of Biotechnology

(1000)

[This question paper contains 4 printed pages.]

04.06.24 (M)

Your Roll No.



Sr. No. of Question Paper : 3095

Unique Paper Code : 32237903

Name of the Paper : Animal Biotechnology

Name of the Course : **B.Sc. (H) Zoology**

Semester : VI (DSE)

Duration : 3 Hours

Maximum Marks : 75

Instructions for Candidates

- Write your Roll No. on the top immediately on receipt of this question paper.
- Answer **five** questions in all.
- Q. No. 1 is compulsory.

1. (a) Define the following : (5)

(i) Genetically modified organism

(ii) Humulin

(iii) DNA Polymerase

(iv) Gene Therapy

(v) Gene Knockout

P.T.O.

3095

2

(b) Differentiate between any **four** of the following :
(8)

- (i) Adaptors and Linkers
- (ii) YAC and MAC
- (iii) Plasmid and M13
- (iv) Southern and Northern Blotting
- (v) Transformation and Transfection

(c) Expand the following : (5)

- (i) BAC
- (ii) NGS
- (iii) SCID
- (iv) SCNT
- (v) RFLP

(d) Write the contributions of the following scientists.
(5)

- (i) Ian Wilmut and Keith Campbell
- (ii) Alec Jeffreys
- (iii) Smith, Nathan and Arber
- (iv) Mario Capecchi & Oliver Smithies
- (v) Boliver and Rodriguez

3095

3

(e) Explain any **two** of the following : (2+2)

- (i) Taq polymerase
- (ii) Transformation efficiency
- (iii) Plaque hybridization

2. (a) Describe type II restriction endonucleases and explain their mode of action. (7)

(b) Describe colony hybridization technique for the screening of genomic library. (5)

3. (a) Describe Western Blotting technique in detail with the help of appropriate diagrams. (7)

(b) Define cloning vectors. Describe the essential features of a cloning vector with suitable examples. (5)

4. (a) Explain the method of producing insect resistant transgenic plants by recombinant DNA technology. (7)

(b) How was 'Dolly the sheep' created? Explain. (5)

5. (a) What is ex-vivo gene therapy? Describe in detail. (7)

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